

# BUILDING DOMAIN LANGUAGE MODELS FROM FIRST PRINCIPLES

How Defense Research Organizations Build and Own  
Domain AI on Workstation Hardware

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## Abstract

Defense research organizations can own domain artificial intelligence today. The ingredients are mathematical literacy, Python proficiency, a curated public domain corpus, and a workstation-class GPU. The Transformer architecture is publicly documented. A complete teaching implementation fits in roughly 500 lines of Python along the nanoGPT lineage.<sup>1</sup> Domain models at ~50 to ~350 million parameters train on a single consumer GPU. The result is AI whose weights, behaviors, and knowledge boundaries remain under organizational control.

This paper gives the construction roadmap and the strategic choices that matter in 2026. Path A builds a domain language model from first principles - maximum auditability and complete ownership. Path B adapts a published open-weight base with continue-pretraining or LoRA / QLoRA inside an air-gapped environment - fastest route to strong domain performance with local inference. Both paths keep queries on hardware you control. That is how command continues when commercial cloud dependency is unacceptable.

*Keywords: domain language model, Transformer, self-attention, tokenization, AdamW, PyTorch, open weights, LoRA, air-gapped training, operational security, auditability*

## 1. Introduction

On 12 June 2017, Vaswani et al. published Attention Is All You Need.<sup>2</sup> Variants of that architecture - GPT, BERT, T5 and their successors - redefined natural-language performance. The public release of the architecture created a lasting opportunity: any research team with mathematical literacy and a modern GPU can implement the core mechanism. Scale of data and engineering investment still separates frontier systems. Domain work - battle-management advisory text, logistics language, technical manuals - runs on a far smaller training budget and still delivers deep subject fluency.

Karpathy's nanoGPT and Let's build GPT from scratch made the teaching stack accessible to every serious engineer.<sup>1,3</sup> This paper takes that foundation into defense research practice: air-gapped operation, full weight ownership, query isolation, and domain depth. The capability is here. The organizations that exercise it own their AI layer.

## 2. Two Paths That Own the Model

### 2.1 Path A - From Scratch (This Paper)

Implement a GPT-style decoder - embeddings, multi-head attention, feed-forward blocks, next-token loss - and pretrain on a curated domain corpus. Teaching and research sizes that fit a workstation: ~3M (nano), ~50M (small), ~350M (medium). Every weight comes from data and choices you control. Path A is the clearest way to understand the machine and the strongest audit trail from corpus to checkpoint.

### 2.2 Path B - Open Weight + Local Adaptation

Bring a published open-weight checkpoint (Llama, Mistral, Qwen and peers) into an air-gapped environment, then continue-pretrain or apply LoRA / QLoRA on domain text.<sup>4,5</sup> Path B delivers strong domain behavior quickly while keeping inference local. Plan the initial ingest as a controlled supply-chain step; after that, the model lives entirely inside your perimeter.

### 2.3 Why Local Ownership Wins

When queries, context, and fine-tuning data stay on organizational hardware, no commercial provider sees the work. Table 6 summarizes the strategic properties of local ownership versus cloud API dependence. For security-sensitive research, local ownership is the architecture that matches the mission.

## 3. Architecture

Sections 3.1-3.4 teach the GPT-style decoder used in nanoGPT: learned positional embeddings, multi-head self-attention, pre-norm LayerNorm, GELU feed-forward, weight tying, and AdamW. Master this stack and you can build Path A end to end.

**Modern open-weight defaults (Path B):** RoPE positional encoding, RMSNorm, SwiGLU feed-forward, and often grouped-query attention. These are the components you will meet when adapting Llama-class weights in 2026 - a natural next step once Path A is fluent.

### 3.1 Token and Positional Embeddings

Each token index maps to a learned vector of dimension  $d_{\text{model}}$ . A position signal supplies order information. Learned absolute positions are the teaching default; RoPE is the modern open-weight default. Weight tying - sharing the token embedding with the output projection - reduces parameter count and improves language-modeling efficiency.<sup>6</sup>

### 3.2 Scaled Dot-Product Attention

$\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k})V$ .<sup>2</sup> Multi-head attention runs this with different projections so the model attends in several representational subspaces at once. Teaching widths commonly use 4-16 heads depending on  $d_{\text{model}}$ .

### 3.3 Transformer Block

Each block combines multi-head self-attention and a position-wise feed-forward network, each with a residual connection and normalization. Pre-norm placement improves training stability. The feed-forward inner width is commonly  $4 \times d_{\text{model}}$  - GELU in the teaching stack, SwiGLU in many modern open models.

### 3.4 Output Projection and Loss

The final block projects to vocabulary size  $V$ . Cross-entropy on the next token drives learning. AdamW with decoupled weight decay is the standard optimizer.<sup>7</sup> Gradient clipping at global norm 1.0 keeps deep training stable. Mixed precision (fp16 or bf16) halves activation memory; on Ampere-class and newer GPUs, bf16 is an excellent default.

**Table 1: Approximate Parameter Count by Configuration**

| Config | $d_{\text{model}}$ | Heads | Layers | FFN   | Params |
|--------|--------------------|-------|--------|-------|--------|
| Nano   | 128                | 4     | 4      | 512   | ~3M    |
| Small  | 512                | 8     | 8      | 2,048 | ~50M   |
| Med.   | 1,024              | 16    | 24     | 4,096 | ~350M  |
| Large  | 2,048              | 16    | 48     | 8,192 | ~2.7B  |

*Polybolos Institute analysis. Nano through Med fit Path A on workstation GPUs; Large scales with institutional hardware.*

## 4. Hardware Capability

A single modern GPU is enough to enter the field. VRAM holds parameters, optimizer states, activations, and the batch. Table 2 gives practical single-GPU bands for Path A full-parameter training with mixed precision and gradient accumulation, and for Path B 4-bit

quantized inference of larger open-weight models.

**Table 2: Hardware Capability Matrix (Path A Train / Path B Infer)**

| Hardware          | VRAM  | Path A train* | Path B infer** |
|-------------------|-------|---------------|----------------|
| RTX 3060          | 12 GB | ~50-100M      | ~7-13B Q4      |
| RTX 4070 Ti       | 12 GB | ~50-100M      | ~7-13B Q4      |
| RTX 4070 Ti Super | 16 GB | ~100-200M     | ~13-20B Q4     |
| RTX 4090          | 24 GB | ~200-350M     | ~13-30B Q4     |
| A100 40GB         | 40 GB | ~1B class     | 30B+ Q4        |
| A100 80GB         | 80 GB | ~2-3B class   | 70B Q4         |

*\*Full-parameter pretrain with mixed precision and gradient accumulation.*

*\*\*4-bit quantized inference with usable context. SKUs are examples; street prices move.*

On A100-class Tensor Cores, FP16/BF16 math throughput reaches up to 16× relative to FP32 - a major reason mixed precision is the modern training default.<sup>8</sup>

## 5. Training Methodology

### 5.1 Data Preparation

Training data quality is the primary lever of model capability at fixed scale.<sup>9</sup> Build the corpus from public, unclassified, and properly licensed sources. Deduplicate, quality-filter, and shuffle. Domain corpora spanning tens to hundreds of millions of tokens give Path A models room to learn specialized language deeply.

### 5.2 Optimization Loop

Sample sequences; forward; cross-entropy; accumulate gradients; backward; clip; AdamW step; zero grads. Cosine decay with linear warmup over the first 5-10% of steps yields stable convergence. Track **microbatch** and **effective batch** (via accumulation) so hardware limits and learning dynamics stay clear.

**Table 3: Suggested Starting Hyperparameters (Path A)**

| Parameter        | Nano ~3M | Small ~50M | Med ~350M |
|------------------|----------|------------|-----------|
| Peak LR          | 3e-3     | 1e-3       | 6e-4      |
| Microbatch       | 4-16     | 2-8        | 1-4       |
| Effective batch* | 64       | 256        | 512       |
| Warmup steps     | 500      | 2,000      | 5,000     |
| Dropout          | 0.1      | 0.1        | 0.1       |
| Seq length       | 256      | 512        | 1,024     |
| VRAM budget      | ~4 GB    | ~8-12 GB   | ~12-24 GB |

*\*Effective batch via gradient accumulation. Starting points - tune on validation loss.*

## 6. Domain Specialization

### 6.1 Corpus Construction

A model trained on domain-relevant text develops specialized representations. For defense research, a public corpus can draw on published doctrine released for public use, technical standards, open acquisition documents, and open literature. Continued pretraining on domain text is a proven adaptation strategy.<sup>10</sup> Curated domain data is how a focused model becomes fluent where general systems stay shallow.

### 6.2 Instruction Tuning

Instruction tuning on (instruction, response) pairs teaches the model to treat input as a command.<sup>11</sup> High-quality sets of hundreds to a few thousand pairs move style and format strongly - LIMA showed how far careful data can go.<sup>12</sup> Pair a solid domain base with sharp instruction data and the system becomes useful to operators and analysts.

### 6.3 Constitutional Alignment

Constitutional AI and preference methods (RLHF, DPO) steer the model toward organizational values and legal constraints.<sup>13,14</sup> Embed rules of engagement language, proportionality, and authority structure into the training and preference loop. Combine that with system prompts, tool gating, and human-on-the-loop authority - a layered design that keeps the model useful and the commander in charge.<sup>15</sup>

**Table 4: Domain Specialization Pipeline**

| Stage       | Objective      | Data                 | Output      |
|-------------|----------------|----------------------|-------------|
| 1. Pretrain | Next-token     | Domain tokens*       | Base LM     |
| 2. Instruct | Follow cmds    | 500-5,000 pairs**    | Instruct LM |
| 3. Align    | Value steer    | Constitution / prefs | Aligned LM  |
| 4. Eval     | Prove strength | Held-out suite       | Report      |

*\*Public / unclassified / licensed corpus for the public recipe. \*\*Quality first; scale with base-model size.*

## 7. Evaluation - Prove the Strength

A serious program records: (1) model size and tokenizer; (2) corpus description and token count; (3) training steps and hardware; (4) validation perplexity; (5) a fixed domain instruction-check set; (6) baselines against untuned open weights and, where policy allows, other systems. Measurement turns a working model into a demonstrable capability - the kind partners and program offices can trust.

## 8. Strategic Implications

### 8.1 Operational Security

Models trained and served inside a controlled environment keep every query, every context window, and every operational phrase on hardware you own. Paths A and B deliver that property by architecture. For organizations that live under OPSEC requirements, local AI is how the mission stays sealed.

### 8.2 Auditability

From-scratch and locally adapted models keep weights and training logs under organizational control. Behavior can be traced. Capability boundaries can be mapped. That audit trail is an institutional asset - full ownership of the AI layer.

### 8.3 Adversarial Robustness

A curated domain corpus and constitutional training raise the bar against casual misuse. Pair them with red-team testing and deployment controls. Extraction research remains an active field; disciplined programs stay ahead of it.<sup>16</sup>

### 8.4 The Proliferation Moment

Open-weight releases and affordable GPUs put domain AI construction within reach of every serious technical organization. Table 5 is an institute judgment of how quickly capable actors can stand up Path B with public tools and active investment. The window to build is open

now.

Table 5: Capability Judgment Matrix

| Factor             | US Defense R&D | Peer State       | Non-State  |
|--------------------|----------------|------------------|------------|
| GPU access         | Advanced       | Advanced / mixed | Commercial |
| Public domain data | High           | High             | Limited    |
| ML expertise       | High           | High             | Variable   |
| Open weights       | Available      | Available        | Available  |
| Path B timeline*   | Weeks-months   | Active           | Months     |

\*Institute judgment; assumes legal access to tools and data.

Table 6: Local Ownership vs Cloud API Dependence

| Property         | Path A/B Local      | Cloud API      |
|------------------|---------------------|----------------|
| Query OPSEC      | Complete isolation  | Vendor-visible |
| Weight ownership | Yes                 | No             |
| Training audit   | A: full / B: strong | Opaque         |
| Domain depth     | High (your corpus)  | General        |
| Supply chain     | Owned after ingest  | Ongoing vendor |
| Ongoing cost     | Electricity         | Per-query fees |

Polybolos Institute analysis.

9. Policy Landscape (2026)

Federal AI direction continues to evolve. Executive Order 14179 (Removing Barriers to American Leadership in Artificial Intelligence) sets the current White House frame following the January 2025 rescission of EO 14110.<sup>17,18</sup> Agency memoranda and the NIST AI Risk Management Framework remain useful engineering references. Build programs that stay current with the OMB and agency line while the technical work moves forward.

10. Recommendations

(1) Stand up Path A on a workstation GPU - teach the architecture and prove auditability. (2) Field Path B for operational domain assistants that must stay air-gapped after ingest. (3) Fund licensed domain corpus pipelines as a first-class research asset. (4) Keep an evaluation card with every released model. (5) Keep human-on-the-loop authority outside the model: language models advise; commanders decide.

11. Conclusion

Building a domain language model from first principles is within reach of any technically capable defense research organization. The architecture is public. The tools are open. The hardware sits on a desk. What remains is the decision to own the AI layer - to train, adapt, evaluate, and serve models whose queries never leave the building.

The cost is engineering time and electricity. The return is operational security, auditability, domain depth, and strategic independence. Assured command needs AI that answers to the organization that fields it. That capability is available now.

Notes

[1] A. Karpathy, nanoGPT, GitHub, 2022. [github.com/karpathy/nanoGPT](https://github.com/karpathy/nanoGPT)  
[2] A. Vaswani et al., Attention Is All You Need, NeurIPS 2017. arXiv:1706.03762

[3] A. Karpathy, Let's build GPT from scratch, YouTube, 2023.  
[4] E. J. Hu et al., LoRA, arXiv:2106.09685, 2021.  
[5] T. Dettmers et al., QLoRA, arXiv:2305.14314, 2023.  
[6] H. Inan et al., Tying Word Vectors and Word Classifiers, ICLR 2017.  
[7] I. Loshchilov and F. Hutter, Decoupled Weight Decay Regularization, ICLR 2019.  
[8] NVIDIA, Ampere Architecture / A100 Tensor Core documentation (FP16/BF16 vs FP32 throughput on A100).  
[9] S. Gunasekar et al., Textbooks Are All You Need, arXiv:2306.11644, 2023.  
[10] S. Gururangan et al., Don't Stop Pretraining, ACL 2020.  
[11] L. Ouyang et al., Training Language Models to Follow Instructions, arXiv:2203.02155, 2022.  
[12] C. Zhou et al., LIMA: Less Is More for Alignment, arXiv:2305.11206, 2023.  
[13] Y. Bai et al., Constitutional AI, arXiv:2212.08073, 2022.  
[14] R. Rafailov et al., Direct Preference Optimization, arXiv:2305.18290, 2023.  
[15] S. Casper et al., Open Problems and Fundamental Limitations of RLHF, arXiv:2307.15217, 2023.  
[16] N. Carlini et al., Extracting Training Data from Large Language Models, USENIX Security 2021.  
[17] EO 14148, Initial Rescissions of Harmful Executive Orders and Actions, Jan 2025 (rescinds EO 14110).  
[18] EO 14179, Removing Barriers to American Leadership in Artificial Intelligence, Jan 2025.

Sources

1. Vaswani et al., Attention Is All You Need. arXiv:1706.03762  
2. Karpathy, nanoGPT / Let's build GPT from scratch.  
3. PyTorch Documentation. [pytorch.org](https://pytorch.org)  
4. Loshchilov and Hutter, AdamW. arXiv:1711.05101  
5. Hu et al., LoRA. arXiv:2106.09685  
6. Dettmers et al., QLoRA. arXiv:2305.14314  
7. Gururangan et al., Don't Stop Pretraining. ACL 2020  
8. Zhou et al., LIMA. arXiv:2305.11206  
9. Bai et al., Constitutional AI. arXiv:2212.08073  
10. Casper et al., Open Problems in RLHF. arXiv:2307.15217  
11. Polybolos Institute. [polybolos.org](https://polybolos.org)

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